

THE WESTERN DESERT: THE THAR

Complete Exercise Solutions Guide

Introduction: This document provides comprehensive, verified, and contextualised solutions to all the textbook activities, in-text explorations, and end-of-chapter exercises featured in the chapter '*The Western Desert: The Thar*'. Every answer is strictly grounded in the official source material to ensure accuracy and educational alignment.

I. In-Text Activities & Warm Up Solutions

Activity: Warm Up! (Page 35)

Task: Circle the words that are related to the desert from the list below:

Polar Bear, Camel, Oasis, Eskimos, Glacier, Cactus, Banjaras, Floods, Grassland.

Solutions & Explanations:

- 1. Camel:** Correct. Known as the 'Ship of the Desert', the camel is the primary mode of transport due to its padded feet and ability to go without food and water for days.
- 2. Oasis:** Correct. An oasis is a place in the desert where underground water comes up to the surface, creating a fertile area where settlements cluster.
- 3. Cactus:** Correct. A typical desert plant that is thorny, with leaves reduced to thorns to prevent excessive loss of water and long roots to tap underground water.
- 4. Banjaras:** Correct. These are the music-loving gypsies/nomads of the Thar desert who act as traders roaming from village to village.

Note: Polar Bear, Eskimos, and Glacier are related to cold, polar regions; Floods and Grassland are typical of wetter, more fertile ecological zones.

Activity: Let's Explore (In-text Research)

1. Find out the name of the hottest state in India (Page 35).

Answer: Rajasthan is the hottest state in India. It contains the Thar Desert, where summer day temperatures are extremely high, frequently exceeding 45°C in cities like Churu and Phalodi.

2. Find out the names of the rivers from which the Indira Gandhi Canal gets its water supply (Page 38).

Answer: The Indira Gandhi Canal gets its water supply from the **Sutlej** and **Beas** rivers, which meet at the Harike Barrage in Punjab, directing water southwest into Rajasthan.

3. Find out the popular traditional dishes of Rajasthan (Page 39).

Answer: The primary traditional dish mentioned in the text is **Daal-Baati and Churma**. Other famous traditional Rajasthani dishes include *Gatte ki Sabzi*, *Ker Sangri*, *Laal Maas*, and *Bajre ki Roti* (millet flatbread).

II. Quick Zone Exercises (Page 41)

A. Tick (✓) the correct options to complete the sentences.

N o.	Question	Options	Correct Answer
1.	The Thar desert is located in the state of ____.	(a) Bihar, (b) Gujarat, (c) Rajasthan, (d) West Bengal	(c) Rajasthan
2.	Oases are found in ____.	(a) mountains, (b) plains, (c) plateaus, (d) deserts	(d) deserts
3.	The Thar Desert is located between ____.	(a) Punjab and Haryana, (b) Rajasthan and Haryana, (c) Maharashtra and Gujarat, (d) Haryana and Gujarat	(d) Haryana and Gujarat (From Kutch to southern Haryana)
4.	____ Canal has changed the life of the people of the desert.	(a) The Jawaharlal Nehru, (b) The Indira Gandhi, (c) The Panama, (d) The Suez	(b) The Indira Gandhi

B. Fill in the blanks.

1. An **oasis** is a fertile place in the desert where water comes up to the surface.
2. The **Aravalli** hills lie to the east of the Thar Desert.
3. Desert plants are usually **thorny** (to prevent excessive loss of water).
4. **Luni** is the only seasonal river in the Thar desert.
5. The nomads of the Thar Desert are called **Banjaras**.

C. Write (T) for true statements and (F) for false statements.

N o.	Statement	Answer	Explanation / Justification
1.	Jaipur is the capital of Rajasthan.	True (T)	Jaipur is the capital and largest city, known as the 'Pink City'.
2.	There is a lot of vegetation in desert region.	False (F)	Due to scanty rainfall, there is very less vegetation, mostly scattered bushes and thorny shrubs.
3.	The sand heats and cools down at a slow speed.	False (F)	Sand heats up very quickly during the day under the sun and cools down very fast at night.
4.	Camels are the most important mode of transport in desert.	True (T)	Camels are perfectly adapted to cross vast sandy terrains due to their long legs and padded feet.
5.	Sand dunes are found in the plains.	False (F)	Sand dunes are hills of sand found specifically in deserts, which keep shifting due to strong winds.

D. Match the columns A and B (Page 42).

Column A	Column B (Correct Match)	Description of Relationship
1. Pink City	(iii) Jaipur	Capital of Rajasthan, famous for its pink-coloured historical buildings.
2. City of lakes	(iv) Udaipur	Famous for its picturesque lakes, palaces, and scenic beauty.
3. Banjaras	(v) Nomads	Gypsies who lead a nomadic life and act as mobile traders in the region.
4. The Aravalli	(ii) Hills	The mountain range that lies to the east of the Thar Desert.
5. Camel	(i) Ship of Desert	The most vital transport, adapted to survive days without food and water.

III. Writing Zone Solutions (Page 42)

Q1. What is a desert?

Answer: A desert is a large area of land which is dry, barren, and receives very little or no rainfall. It is often covered with sand and described as desolate. Because of the scarce water availability, very few plants and trees can survive, and the landscape has very little vegetation to hold the soil in place.

Q2. Why is it very hot in the desert?

Answer: The desert is extremely hot during the day because sand heats up very quickly under direct sunlight. Due to the lack of cloud cover, dense forest vegetation, or water bodies, solar radiation directly strikes the ground. Conversely, the nights are cool because sand also cools down very quickly once the sun sets, creating a wide temperature range.

Q3. Why is camel the most suitable transport for desert region?

Answer: The camel is the most suitable mode of transport in the desert due to several physiological adaptations:

- **Long Legs and Padded Feet:** They prevent the camel from sinking into the loose, deep desert sand and allow easy movement.
- **Hump:** The hump is a mound of fat that provides energy and allows the camel to travel long distances without food and water for many days.
- **Thick Skin and Retaining Water:** It is highly adapted to withstand high temperatures and conserve body fluids.

Q4. What are the characteristics of the Thar vegetation?

Answer: The vegetation of the Thar Desert has adapted to the extremely dry climate and scanty rainfall in the following ways:

- **Thorny Shrubs:** Scattered bushes and thorny shrubs (like kikar, babul, khair, and cactus) require very little water to grow.
- **Leaves Reduced to Thorns:** Thorny plants have leaves modified into thorns to prevent excessive water loss through transpiration.
- **Long Roots:** Trees have long roots that penetrate deep into the ground to reach underground water tables.
- **Thick Barks:** Thick barks protect the trunk and prevent moisture evaporation.

Q5. How is the Indira Gandhi Canal of great significance to the people of Rajasthan?

Answer: The Indira Gandhi Canal (or Rajasthan Canal) has transformed life in the Thar Desert by:

- **Mitigating Scarcity:** It brings fresh river water from the Sutlej and Beas to dry, water-scarce areas, relieving people of walking miles to fetch domestic water.
- **Agricultural Expansion:** It has brought vast arid tracts under cultivation, enabling farmers to grow crops like bajra, jowar, wheat, and mustard.
- **Socio-economic Improvement:** It provides a reliable water and electricity supply to villages, stabilizing settlements and easing daily survival.

IV. Thinking & Application Zones (Page 42)

Problem Solving Zone

Task: In desert region, there is always shortage of water. People have to walk miles to collect water. Suggest few ways by which this shortage of water could be tackled.

Water Scarcity Management Plans

Proposed Solutions:

1. **Rainwater Harvesting (Tanks/Tankas):** As highlighted in the text, households can construct underground storage tanks to collect and store every drop of rain received during the brief wet spell. This serves as an immediate, local source during dry seasons.
2. **Canal Irrigation Systems:** Expanding and maintaining distribution channels from the Indira Gandhi Canal ensures that remote desert villages receive water for both domestic and agricultural use.
3. **Drip and Sprinkler Irrigation:** Moving away from traditional flood irrigation to highly targeted drip systems minimizes evaporation losses and conserves groundwater resources.
4. **Afforestation (Green Belts):** Planting desert-adapted, drought-resistant trees (like khair and babul) helps stabilize sand dunes, prevent soil erosion, retain ground moisture, and recharge local aquifers.
5. **Desalination and Water Treatment:** Installing small-scale solar desalination plants to purify brackish underground water can supply safe drinking water locally without relying on distant sources.

Life Skills Zone

Task: Imagine that you are stranded on a deserted island. What are the things would you need in that situation? What would you do to survive? Make a list of five things that you would do to survive on the deserted island till the rescue team comes.

Wilderness Survival Protocol

Survival Checklist and Action Plan:

Essential Needs: Fresh drinking water, temporary protective shelter, fire-starting material, signaling tools, and basic foraging tools.

Top 5 Survival Actions:

1. **Secure Fresh Water:** Immediately search for fresh water (rainwater, streams, or coconuts) to avoid life-threatening dehydration.
2. **Construct a Protective Shelter:** Build a lean-to or a leaf shelter using palm branches and wood to protect against extreme sun, wind, and wild animals.
3. **Build and Maintain a Fire:** Start a fire using dry wood and friction or sparks. Fire provides warmth, deters predators, purifies water, and cooks food.
4. **Create Visible Rescue Signals:** Lay out a large 'SOS' sign using dark stones or logs on the sandy beach, and keep dry wood ready to ignite a smoke signal if a plane or ship is spotted.
5. **Forage for Safe Food:** Look for easily accessible food sources like tropical fruits, crabs, or fish while avoiding unknown berries to prevent poisoning.

Curiosity Zone

Task: The nomads in the deserts move from one place to another with their families. Do you ever wonder how their houses would be like and how they might have prepared or cooked food? Think and write.

Analysis and Insight:

- **Nomadic Housing (Tents/Khabas):** Since desert nomads (like the Banjaras) are constantly on the move in search of water and pasture, their shelters must be lightweight and highly portable. They use temporary tents crafted from woven goat hair, heavy canvas, or local bamboo poles, which can be easily packed onto camels or bullock carts within minutes. In semi-permanent camps, they might build simple mud-walled structures with conical thatched roofs made of dry thorny desert grass, which keeps the interior cool during the day.
- **Nomadic Cooking Techniques:** Nomads use portable, semi-circular clay stoves (called *chulhas*) built directly on the ground. For fuel, they gather dry thorny twigs, desert shrub wood, or dried animal dung cakes. Because water is scarce, their culinary techniques prioritize dry roasting, baking over embers (such as making thick millet flatbreads directly on coals), and cooking foods that have a long shelf life (like dried lentils, beans, and clarified butter/ghee) that do not require refrigeration.